

DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING

B.TECH DEGREE II INTERNAL EXAMINATION July 2024

Semester IV CS 1404 AUTOMATA LANGUAGES AND COMPUTATION

Faculty: PRAMOD PAVITHRAN

Time: 2Hrs

Marks: 50

CO 2 Analyse and generate regular expressions for any structure.

CO 3 Demonstrate that a given language is regular or not.

CO 4. Design new context free grammar.

Part A (Answer All Questions)

(5 x 4 = 20)

- I. a. Convert the following regular grammar to a Finite Automata
S $\rightarrow$ 0A | 1B
A $\rightarrow$ 0A | 1B | 0
B $\rightarrow$ 1A | 0B | 1
CO2
- b. Show how left recursion is eliminated in the following rules:
(a) A $\rightarrow$ Abc | bB | a
(b) B $\rightarrow$ Baa | Bbb | c
CO4
- c. Show that the grammar
E $\rightarrow$ E + E | E - E | id is ambiguous
CO4
- d. Construct the string baaabbba from the grammar
S $\rightarrow$ AB
A $\rightarrow$ 0A | 1B | 0
B $\rightarrow$ 1A | 0B | 1 by using
(i) left most derivation
(ii) right most derivation
CO4
- e. Find the parse tree for generating the string abaabaa from the given grammar
S $\rightarrow$ aAS | a
A $\rightarrow$ bS
CO4

Part B (Answer Any Three Questions)

(3 x 10 = 30)

- II. a. Convert the following grammar to Greibach normal form.
S $\rightarrow$ Abb | a
A $\rightarrow$ aaA | B
B $\rightarrow$ bAb
CO4 5
- b. Convert the following grammar to CNF.
S $\rightarrow$ abA | caB
A $\rightarrow$ bAB | aB | aA | b
B $\rightarrow$ AASB | bSA | b | ϵ
CO4 5

- III. State the pumping lemma for regular sets. Using the lemma show that
CO3 10

$L(G) = \{ a^n b^n \mid n > 0 \}$ is not a regular language

IV. Write a CFG for the language CO4 10

(i) $L(G) = \{ a^n b^m c^m d^{2n} \mid n > 0, m > 0 \}$

(ii) $L(G) = \{ a^n b^n \mid n > 0 \}$

(iii) $r = (a+b)^* cc (a+b)^*$

(iv) Positive integers with one, two or three digits

V. a Remove useless symbols from the following grammar CO4

$S \rightarrow aB \mid bX$

$A \rightarrow BAd \mid bSX \mid a$

$B \rightarrow aSB \mid bBX$

$X \rightarrow SBD \mid aBx \mid ad$

5

b Remove unit productions from the following grammar CO 4 5

$S \rightarrow aB \mid Xa$

$A \rightarrow Bd \mid bSX \mid X$

$B \rightarrow aB \mid bBX \mid C$

$C \rightarrow a$

$X \rightarrow SBD \mid aBx \mid ad$